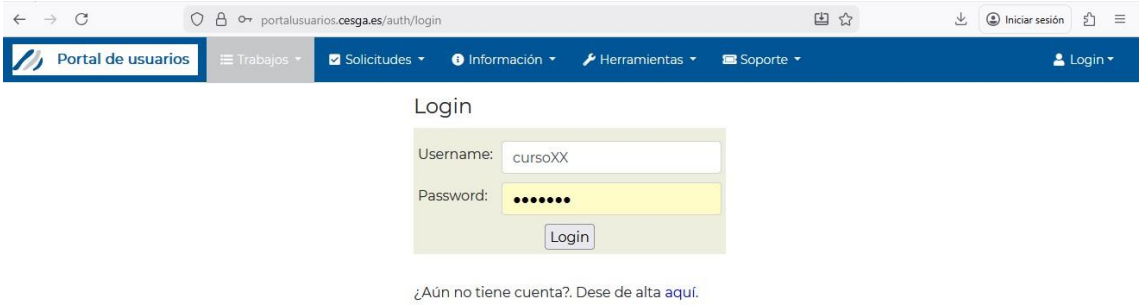
STEPS FOR CONNECTING QMIO

1. Access the CESGA User Portal

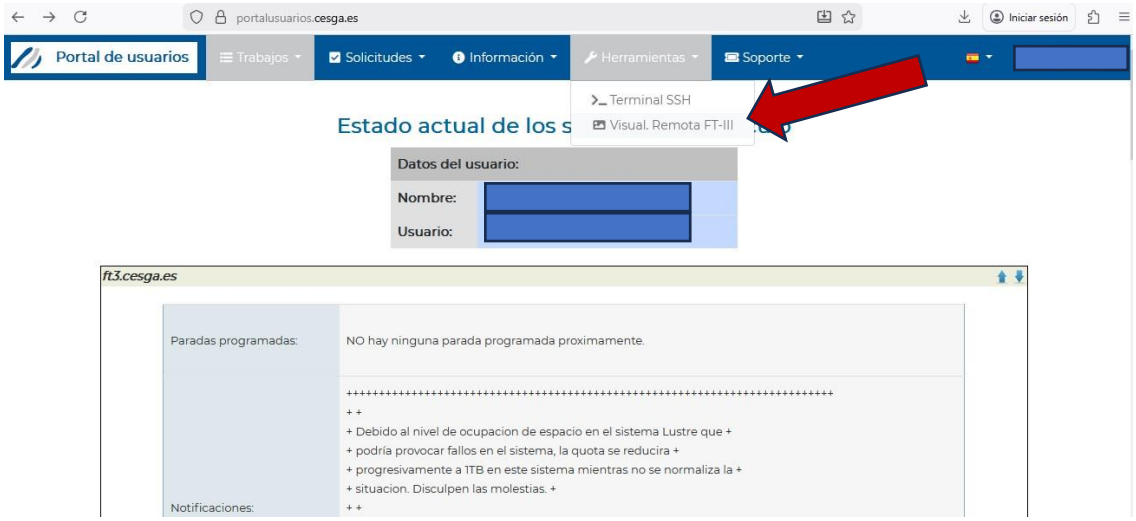
https://portalusuarios.cesga.es/auth

BY DEFAULT, THE INTERFACE IS IN SPANISH. YOU CAN CHAGE TO ENGLISH VERSION USING THE FLAGS ON THE RIGHT

2. Log in using your assigned **username and password**.



3. In the top menu, go to:
   Tools → Remote Visualization (FT-III)



Click the **large blue button** to create a new remote desktop session.

Accept the pop-up confirmation (in Spanish).

THIS WILL SHOW A NEW SCREEN LIKE

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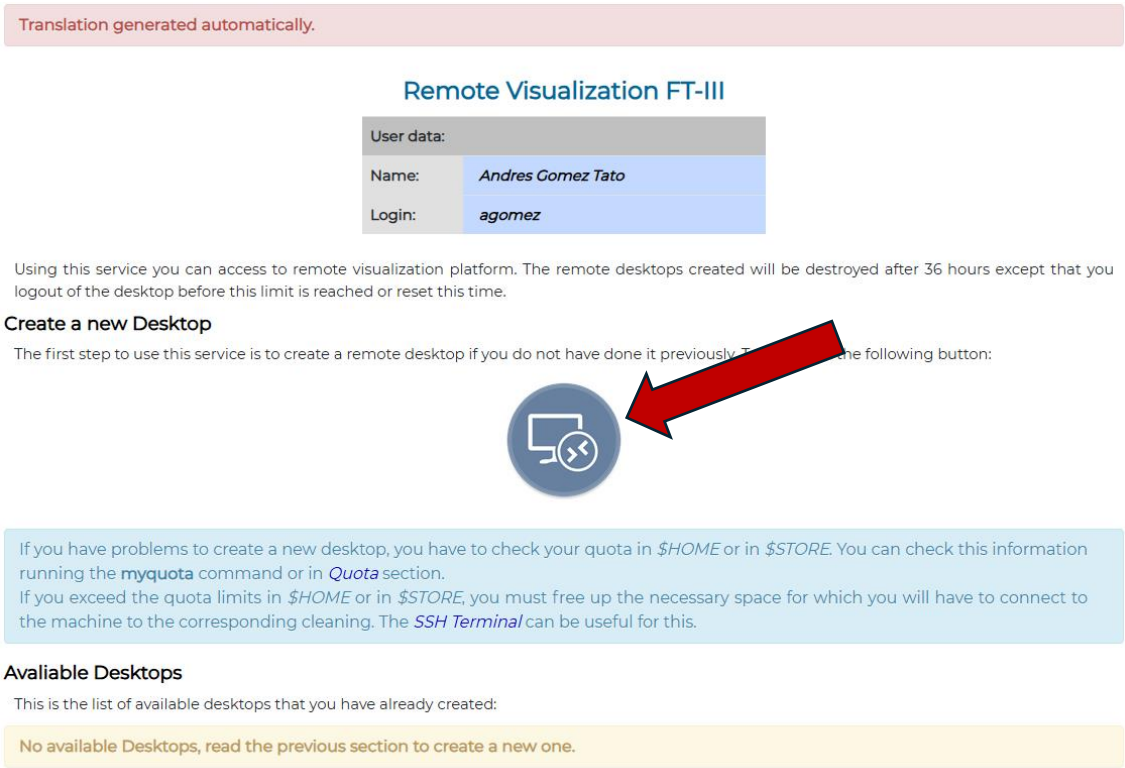
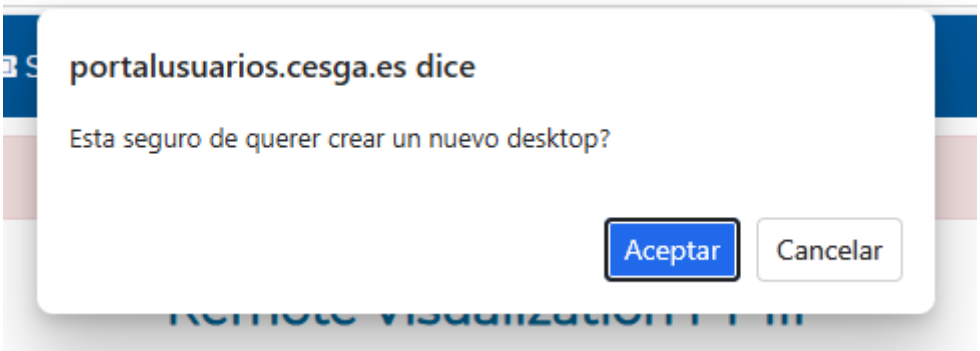
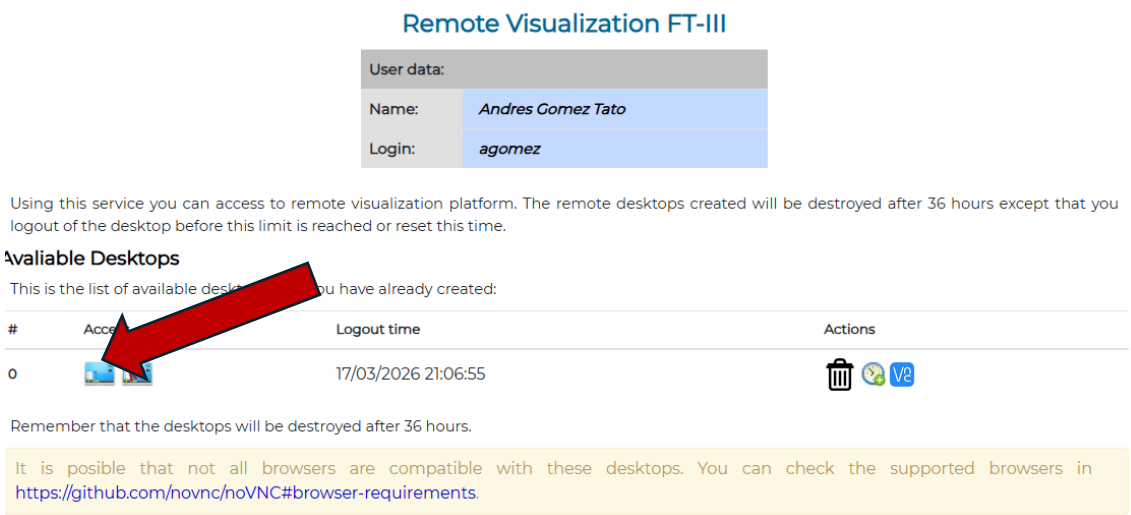


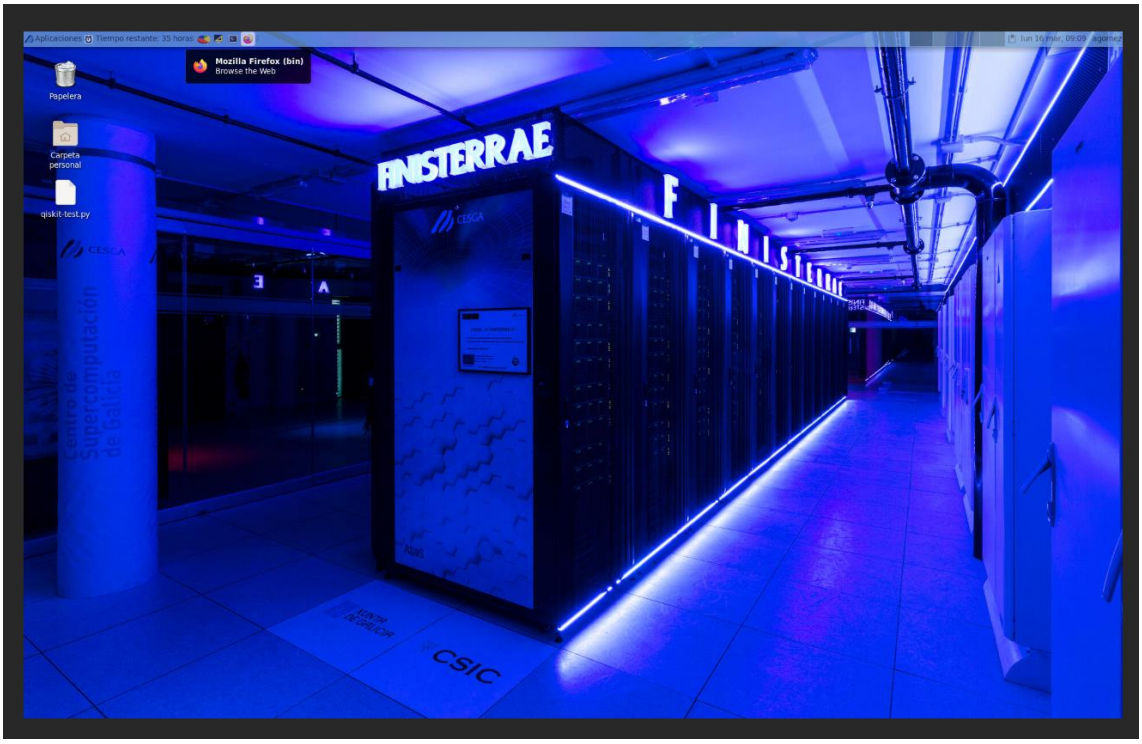
4. PUSH THE BIG BLUE BUTTON IN THE MIDDLE OF THE SCREEN. ACCEPT THE MESSAGE IN THE POP-UP (IN SPANISH)



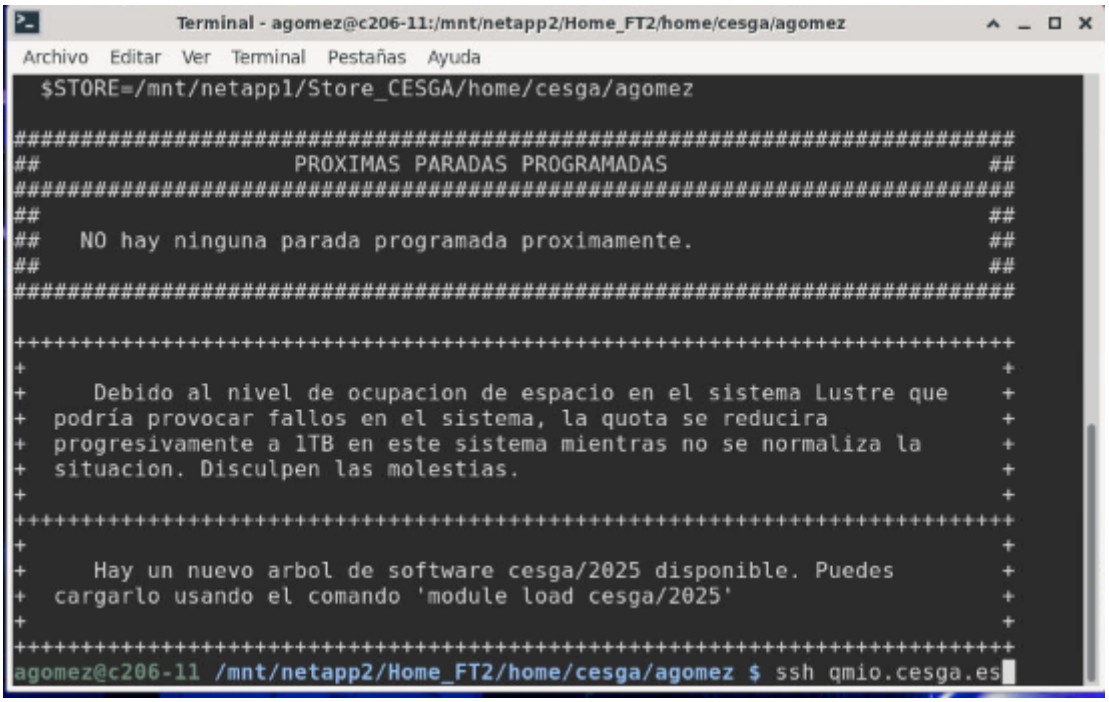
5. AFTER A COUPLE OF SCREENS, YOU WILL SEE A SCREEN LIKE THIS:



6. THE LEFT BLUE MINI-SCREEN BUTTON WILL OPEN YOUR VIRTUAL DESKTOP. THE NEXT ONE ON THE LEFT PERMITS TO SHARE A READ-ONLY SCREEN.



7. PUSH THE ICON TO OPEN A TERMINAL
8. NOW YOU CAN CONNECT TO QMIO USING A SSH COMMAND
   ssh qmio.cesga.es
   You are now connected to the QMIO infrastructure.



INFO ABOUT QMIO

1. IN THE PORTAL https://cesga-docs.gitlab.io/qmio-user-guide/index.html SELECT INFORMATION/LINKS
2. THERE, YOU WILL FIND INFORMATION ABOUT QMIO
3. CESGA DOES NOT PUBLISH IN THE WEBPAGE INFO ABOUT THE CALIBRATION OF THE QPU. THIS CAN BE GET FROM THE QMIO BACKEND IN QISKIT. ALSO, WHEN YOU LOAD QMIOTOOLS, YOU WILL HAVE THE NEXT VARIABLE IN THE ENVIRONMENT WITH THE FOLDER OF THE CALIBRATIONS:

   QMIO_CALIBRATIONS=/opt/cesga/qmio/hpc/calibrations

   The folder can change. So, please, use the variable. There, there is one JSON file per day of calibration. Be aware that sometimes the T2=0. This is due to errors in the weekly benchmarking (T2 is measured only once per week). This can produce inaccurate solutions in the FakeQmio backend if the used qubits include this faulty qubits.
   You can also access to the calibration information using Qmiotools. Using qiskit backend

   ```
   from qmiotools.integrations.qiskitqmio import QmioBackend
   Backend=QmioBackend()
   #Print instructions:
   print(Backend.target)
   #Print Qubit information
   print(Backend.target.qubit_properties)
   ```

In this kink you can find some examples, and you can follow the instructions below to use them

Example Workflows on QMIO

You can run example using:

bash start_current.sh

example Benasque.ipynb

INFO ABOUT QULACS Emulator

IN THE PORTAL https://cesga-docs.gitlab.io/qmio-user-guide/index.html SELECT INFORMATION/LINKS. THERE, YOU WILL FIND INFORMATION ABOUT.

QULACS is a **high-performance quantum simulator** available on CESGA HPC infrastructure, optimized for:

- Large-scale simulations
- Execution on **A64FX nodes**
- MPI parallelization

You can access the A64FX partition in two ways:

Batch job submission:

sbatch -p a64 --mem-per-cpu=1G --time=10:00:00 job.sh

Interactive session:

compute --a64

Load Required Module

module load qulacs-hpcx (this module only works on the ARM nodes)

In this kink you can find some examples, and you can follow the instructions below to use them

Example Workflows on qulacs

You can run simulations using:

- **Single-node execution**

  bernstein_vazirani.py

- **MPI parallel execution**

  bernstein_vazirani_mpi.py

- **Submission script example**

sbatch submit_DVBANC_.sh

(submit_DVBANC_.sh is a provided script to compare the performance of QULACS simulations with and without MPI.)

Hackathon Reservation Details

The following reservations have been configured for your hackathon:

ReservationName=Benasque_QPU
StartTime=2026-03-19T17:00:00
EndTime=2026-03-20T06:00:00

ReservationName=Benasque_ILK
StartTime=2026-03-19T12:00:00
EndTime=2026-03-20T06:00:00
Nodes=c7-[7,13,19-22]

ReservationName=Benasque_ARM
StartTime=2026-03-19T12:00:00
EndTime=2026-03-20T06:00:00
Nodes=c7-[101-116]

*****Please note that your account will be deactivated after the end of the hackathon.**
We kindly ask you to upload your project to this repository to ensure it is properly preserved: https://cumulo.cesga.es/index.php/s/wkg8RebzAgLJda3